CLASS 10 · CBSE · NCERT MATHEMATICS · DPP SHEET

Chapter 9: Some Applications of Trigonometry

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. A tower stands on level ground. From a point 45 m from its foot, the angle of elevation of its top is 30°. Find the height of the tower.

2. A kite is flying at a height of 75 m above the ground. Its string makes an angle of 60° with the ground. Find the length of the string, assuming there is no slack.

3. A pole 9 m high casts a shadow $3\sqrt{3}$ m long. Find the sun's angle of elevation.

4. A tree breaks and its top touches the ground 12 m from its foot, making an angle of 30° with the ground. Find the original height of the tree.

5. From the top of a cliff 75 m high, the angle of depression of a boat is 60°. Find the distance of the boat from the foot of the cliff.

6. An observer 1.5 m tall is 20.5 m from a tower 22 m high. Find the angle of elevation of the top of the tower from his eyes.

7. From a point on the ground, the angles of elevation of the bottom and the top of a flagstaff fixed on a building 20 m high are 45° and 60°. Find the height of the flagstaff.

8. Two poles of equal height stand on opposite sides of a road 100 m wide. From a point on the road between them, the angles of elevation of their tops are 60° and 30°. Find the height of the poles and the distances of the point from them.

9. From the top of a 12 m high building, the angle of elevation of the top of a tower is 60° and the angle of depression of its foot is 45°. Find the height of the tower.

10. The angle of elevation of the top of a tower from a point on the ground is 30°. On walking 20 m towards the tower, the angle becomes 60°. Find the height of the tower.

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Answer Key

1. $\frac{45}{\sqrt{3}} = 15\sqrt{3} \approx 25.98$ m
2. $\frac{75}{\sin 60^\circ} = 50\sqrt{3} \approx 86.6$ m
3. $\tan\theta = \frac{9}{3\sqrt{3}} = \sqrt{3}$, so $\theta = 60^\circ$
4. $\frac{12}{\sqrt{3}} + \frac{24}{\sqrt{3}} = 12\sqrt{3} \approx 20.78$ m
5. $\frac{75}{\sqrt{3}} = 25\sqrt{3} \approx 43.3$ m
6. $\tan\theta = \frac{20.5}{20.5} = 1$, so $\theta = 45^\circ$
7. $20\sqrt{3} - 20 = 20(\sqrt{3} - 1) \approx 14.64$ m
8. Height $25\sqrt{3} \approx 43.3$ m; distances 25 m and 75 m
9. $12(1 + \sqrt{3}) \approx 32.78$ m
10. $10\sqrt{3} \approx 17.32$ m

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